Fertilizer Programme 2026/27.

Fertilise Smarter, Grow Stronger.

Establishment-year nutrient strategy for a 14 ha mixed stand of barley, oats and lucerne under centre-pivot irrigation across the Kraan and Gamka blocks.

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SECTION 01

Background.

The plan for the 2026/27 season is to sow a mixed stand of barley, oats and lucerne under centre-pivot irrigation across 14 hectares spanning the Kraan and Gamka blocks. The cereals come off as forage hay in August; the lucerne takes over thereafter as a multi-year perennial stand.

This document sets out the nutrient strategy for the establishment year and flags the priorities for Year 2 and beyond.

The practice of nurse-cropping lucerne with cereals under irrigation is agronomically sound in the Karoo. Recent South African research has cautioned against nurse cropping in the dryland Mediterranean winter-rainfall zone due to moisture competition — but the Prince Albert irrigated context removes that concern. The cereals deliver Year-1 ground cover and a harvestable hay cut while the lucerne establishes its taproot and rhizobial nodulation underneath.

PROGRAMME AT A GLANCE

- 14 ha under centre-pivot irrigation across the Kraan and Gamka blocks
- Mixed stand: barley + oats (Year-1 hay cut) nurse-cropping lucerne (multi-year perennial)
- Three on-pellet passes across the season plus targeted synthetic topdresses
- Post-harvest soil re-test built into the plan before any Y2 decisions are made

SECTION 02

Reading of the Soil.

The April soil analysis tells a coherent story across both blocks. For the purpose of this programme the two are treated as one averaged field.

2.1 What's already in good shape

- **Calcium** is abundant on both blocks — no lime needed. Unusual for South African soils and a real asset.
- **Magnesium** is adequate.
- **Ca:Mg ratio sits at 5.3 : 1** — almost exactly the target for lucerne (ideal 5 : 1).
- **Sulphur** is already Very High in the soil exchange.
- **Potassium** currently classifies as Optimal.
- **Acid saturation is effectively zero** — the opposite problem is the concern here.

2.2 What needs attention

PARAMETER	READING	TARGET	STATUS
Phosphorus (Olsen)	Low	Moderate–High	LOW
Boron	~1.0 mg/kg	1.5–2.5 mg/kg (lucerne)	LOW
Sodium — Gamka	14.7% base saturation	< 5%	SODIC THRESHOLD
Sodium — Kraan	8.7% base saturation	< 5%	ELEVATED
Organic matter	~1.1% (avg.)	≥ 2% for perennial lucerne	LOW
Soil pH (KCl)	7.6 – 8.3	5.8 – 6.8 (lucerne)	ABOVE OPTIMUM

The Bray II phosphorus values on the report are inflated by the alkaline carbonate chemistry and should be disregarded — the Olsen figure is the reliable one at this pH. Acidification is not being attempted this season; the Karoo's carbonate buffering makes that a multi-season, high-cost exercise without a guaranteed result. The question gets revisited once Year 1 performance is visible.

2.3 The three loudest signals

Sodium is the biggest long-term soil-health concern.
Gamka at 14.7% base saturation sits right at the sodic threshold. Unaddressed, structure degrades and the lucerne stand will not carry across years.

Boron is limiting for lucerne.
Sitting around 1.0 mg/kg against a 1.5–2.5 target. Drives yellowing and poor regrowth if not corrected.

Organic matter is too low to buffer a productive multi-year stand.
Building soil carbon is the most important multi-year investment on this farm.

SECTION 03

The Strategy.

The programme serves three objectives, each on a different timescale.

(A) SHORT TERM	**Carry the cereals through to a good hay cut in August.** Moderate nitrogen, adequate phosphorus and potassium — drawing heavily on what's already in the soil and the product on hand.
(B) MEDIUM TERM	**Establish a viable lucerne stand underneath.** Keep nitrogen deliberately modest. Heavy N suppresses rhizobial nodulation — the lucerne draws from the cereal's N and never learns to fix its own. Synthetic N is held back and the legume makes up the balance from Year 2 onward.
(C) LONG TERM	**Build organic matter and displace sodium** — uniformly, continuously, across the whole season. This is where the **Rescue + Gypsum** pellet earns its place. One product delivers organic carbon, slow-release nitrogen, a full micronutrient package (boron, zinc, manganese, copper), calcium, sulphur, and gypsum for sodium displacement — all in a single pass. No separate boron broadcast, no standalone gypsum truck, no lime.

SECTION 04

Applications.

4.1 Planting (Tuesday)

PLANTING · 10 ha + 4 ha (starter already down) Day 0 Tuesday

10 HA BEING PLANTED NOW

Apply the **21% N + 24% S granule** at **100 kg/ha** — delivering **21 kg N + 24 kg S per hectare**. The nitrogen is in the mixed-stand sweet spot: enough for cereal vigour, modest enough to allow lucerne nodulation. The sulphur rides along as a carried-forward redundancy (soil S is already surplus), which is fine because the product is on hand and earns its place on the nitrogen.

4 HA ALREADY PLANTED WITH STARTER

Carrying **2:3:4 (30)** at 200 kg/ha — **13 kg N + 20 kg P + 27 kg K per hectare**. A good establishment profile, slightly P- and K-heavier than the 10 ha but light on N. The nitrogen evens out at week 2.

4.2 Week 2 — Rescue + Gypsum pellet at 800 kg/ha (all 14 ha)

WEEK 2 · RESCUE + GYPSUM PELLET · 800 KG/HA Day 14 ~12 May

PER HECTARE, THIS SINGLE PASS DELIVERS

COMPONENT	PER HA THIS PASS	NOTE
Nitrogen (total)	~13 kg	About 20–30% plant-available this season; the rest mineralising across Y2 + Y3
Phosphorus	~6 kg	
Potassium	~10 kg	
Calcium	~54 kg	
Magnesium	~6 kg	
Sulphur	~35 kg	
Boron	**0.4 kg**	**The one that matters for lucerne**
Zinc, manganese, copper	~0.35 kg total	Trickled
Organic carbon	**~220 kg**	
Gypsum (Na displacement)	~170 kg	

ON THE 4 HA ONLY — SIDE-DRESS

Add **50 kg/ha of the 21% N + 24% S granule** — delivers 10.5 kg N + 12 kg S per hectare, bringing the 4 ha in line with the 10 ha on starter nitrogen.

4.3 Week 6 — second pellet pass + synthetic topdress (all 14 ha)

WEEK 6 · PELLET + SYNTHETIC TOPDRESS Day 42 Cereals tillering

INPUT	RATE	PER-HA DELIVERY
Rescue + Gypsum pellet	800 kg/ha	Same delivery profile as week 2
4:1:1 (31)	25 kg/ha	5.2 N · 1.3 P · 1.3 K
3:1:5 (36)	28.6 kg/ha	3.4 N · 1.1 P · 5.7 K

Combined synthetic topdress delivers → ~9 kg N · ~2 kg P · ~7 kg K per hectare

Uses every kilogram of both on-hand blends — nothing left over, nothing wasted. This is the "cereals are tillering, give them something" pass.

4.4 After August hay cut — third pellet pass (all 14 ha)

POST-HARVEST · RESCUE + GYPSUM PELLET · 800 KG/HA Aug Cereals off · Lucerne takes over

At this point the cereals are off, the lucerne takes over, and the perennial phase begins. The pellet carries everything the young lucerne stand needs to push through the first real productive cuts.

BUILT-IN DECISION POINT — POST-HARVEST SOIL RE-TEST

A cereal hay cut removes roughly **100 kg of potassium per hectare** — meaningfully more than will have been applied.

If the re-test shows soil K dropped from Optimal to Low, top up with approximately **150 kg/ha of a 60% K_2O source**. That's a decision made against real data, not speculation. If K held, skip it.

SECTION 06

Why the Pellet Works Over Multiple Years.

Rescue + Gypsum is doing three different jobs on three different timescales.

MONTH 1	The **gypsum component dissolves** and begins displacing sodium off the cation exchange. Each pass delivers about 170 kg/ha of gypsum, so the three passes across the season total just over **500 kg/ha**. Not an aggressive rate, but consistent across 14 weeks of active soil biology under pivot moisture. By August the exchangeable sodium on Gamka should have moved from roughly 15% of base saturation down toward 11–12%. *Full clearance below the sodic threshold needs 2–3 seasons of this.*
SEASON 1	About **20–30% of the pellet's nitrogen mineralises** and becomes plant-available, along with the boron, zinc, manganese and copper. That's precisely why no separate boron broadcast is needed — the pellet is covering it. The remaining micronutrients work through the season at soil-biology pace.
YEARS 2–4	The remaining **70% of the pellet's nitrogen** releases gradually as soil biology continues mineralising the organic fraction. Carbon accumulates. Phosphorus — which is otherwise rapidly locked up by this soil's high calcium as unavailable calcium phosphates — is held in organic forms by the manure carrier and released progressively through microbial turnover. This is the core reason a repeated organic programme builds productivity in a way that synthetic-only programmes cannot: it's not just feeding this crop, it's building the soil's carrying capacity for every future crop.

CARBON — THE REAL INVESTMENT

At the current soil organic carbon of 0.4–1.5% across the two blocks, a **660 kg C/ha** contribution this season is genuinely visible on next year's lab report. Expect organic matter to climb from ~1.1% toward **1.5–1.8% by this time next year**, and to continue climbing if the programme is repeated annually.

A lucerne stand depends on soil structure and biology as much as it does on chemistry. Calcareous Karoo sand at 1% OM will never carry the same stand as the same mineralogy at 2.5% OM.

SECTION 07

Nutrient Balance — In vs Out.

Per hectare, averaged across the full programme against the expected August hay removal.

NUTRIENT	APPLIED Y1 (PLANT-AVAILABLE)	REMOVED IN HAY CUT	NET
Nitrogen	~30–40 kg	~100 kg	Negative; lucerne fixation recovers this from Y2
Phosphorus	~22 kg	~12 kg	Slightly positive ✓
Potassium	~38 kg	~100 kg	Negative; drives the conditional post-August top-up
Calcium	~160 kg	~15 kg	Strongly positive ✓
Magnesium	~19 kg	~5 kg	Positive ✓
Sulphur	~130 kg	~10 kg	Strongly positive (soil already surplus)
Boron	~1.2 kg	trace	Positive ✓
Organic carbon	**~660 kg**	0	**The real long-term investment**

SECTION 08

Ratios — Where You Are and Where We're Heading.

RATIO / SATURATION	CURRENT	TARGET	STATUS
Ca : Mg	5.3 : 1	5 : 1	IDEAL — NO INTERVENTION
(Ca + Mg) : K	~37 : 1	20–30 : 1	K RELATIVELY UNDER-REPRESENTED
Sodium saturation — Kraan	8.7%	< 5% (3-yr)	ELEVATED
Sodium saturation — Gamka	14.7%	< 5% (3-yr)	SODIC THRESHOLD
Acid saturation	~0%	< 10%	NOT A CONCERN

HOW WE'RE MOVING THE NUMBERS

- **Ca:Mg** essentially ideal — no intervention.
- **(Ca+Mg):K** sits high of target for hay/forage; says K is relatively under-represented and will be tracked via the post-August re-test.
- **Sodium** — the pellet's gypsum component drives this down across three seasons.
- **Acid saturation** is effectively zero; the opposite (carbonate over-buffering) is the issue and is revisited from Y3.

SECTION 09

Year 2 and Beyond.

This is the hardest year because it's establishment on ground that isn't naturally favourable to lucerne. Realistic expectations:

YEAR 1 — Establishment

- Cereals cut August — roughly **5 tons of mixed hay per hectare**.
- Lucerne stand establishes but will be thin under the cereal. Visible after the hay cut, not yet productive.

YEAR 2 — First real lucerne cuts

- First real lucerne cuts from **December 2026** onward. Realistic range **6–10 tons hay/ha** across the season.
- Soil re-test in February tells us whether the pH, sodium saturation, organic matter and boron have moved as expected.

YEAR 3 — Productive stand

- Target lucerne productivity **12–15 tons hay/ha**.
- Target soil organic matter ≥ 2%.
- Target sodium saturation < 10% on Gamka, < 5% on Kraan.
- Target boron into the lucerne-optimal 1.5–2.5 mg/kg band.
- Phosphorus building from Low toward Optimal.

YEAR 3–4 DECISION POINT

If pH hasn't moved and the stand is underperforming relative to the above, **elemental sulphur becomes economical at that stage** — not sooner. The Karoo's carbonate buffering will swallow earlier attempts.

The on-going programme to get there is straightforward: **continue the Rescue + Gypsum pellet at similar cadence annually** — this is the engine — and top up phosphorus and potassium each season against real post-harvest soil data, rather than speculation.

SECTION 10

Items We're Carrying Into The Season.

For transparency, these are the open questions in the programme — places where the best call has been made with the information available, but there are variables worth tracking.

1. **Pivot water quality is untested.** This is the biggest unknown in the programme. If the water carries significant sodium or bicarbonate, the gypsum is effectively treading water. A water sample sent to the lab this season lets next year's gypsum rate be sized properly against a known result.
2. **Potassium balance** — the hay cut will draw soil K down below optimum. The top-up is scheduled as a conditional post-August decision based on the soil re-test.
3. **Soil pH sits above lucerne's optimum on both blocks.** Expect a thin Y1 lucerne stand. Year-2 performance tells us whether we're on an acceptable long-term path without major intervention.
4. **Gamka vs Kraan gypsum need** — Gamka ideally wants about twice the rate of Kraan. The pellet format and single-truck constraint made uniform treatment the pragmatic choice this year; future seasons can split the rate by block.

END OF PROGRAMME

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Application rates assume uniform spreader calibration and are given on a weighted-average basis across the two blocks.